



Customer Shopping Behaviour Analysis

An end-to-end data analytics project uncovering purchasing patterns, demographic trends, and revenue opportunities from 3,900 retail transactions.

PROJECT REPORT

Project Overview & Problem Statement



Objective

Help a retail company understand customer purchasing behaviour to improve sales, satisfaction, and long-term loyalty.

Key Focus Areas

- Customer spending patterns
- Impact of discounts and promotions
- Product category performance
- Subscription behaviour
- Demographic-based purchasing trends

Dataset & Preprocessing

Dataset Summary

3,900 rows × 18 columns covering demographics, purchase details, behavioural attributes, and logistics.

Missing Data

Review Rating had **37 missing values** — imputed using **median per category**. Redundant column `promo_code_used` removed.

Demographics

Age, Gender, Location,
Subscription Status

Purchase Details

Item, Category, Amount,
Season, Size, Colour

Behavioural

Discount Applied, Review Rating, Purchase Frequency

Feature Engineering & Database Integration



New features — **Age Group** (quantile-based binning) and **Purchase Frequency Days** (categorical-to-numeric conversion) — enriched the dataset before loading into PostgreSQL via SQLAlchemy for structured business querying.

Revenue & Spending Insights

\$157.9K

Male Revenue

vs. \$75.2K from female customers

\$60.48

Express Avg. Spend

vs. \$58.46 for Standard shipping

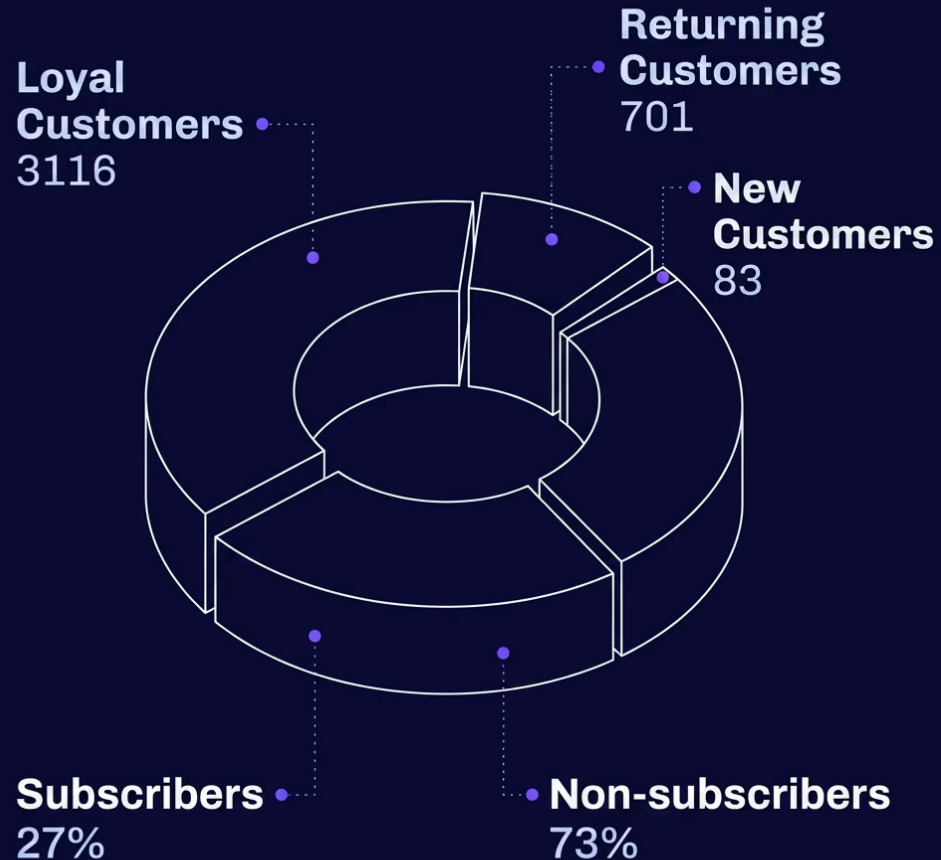
\$59.76

Overall Avg. Purchase

Reflecting mid-range spending
behaviour

Express shipping customers spend slightly more on average, suggesting a correlation between urgency and higher purchase value.

Subscription & Customer Segmentation



Subscription Gap

Only **27%** of customers are subscribers — a major growth opportunity. Avg. spend is nearly equal (\$59.49 vs \$59.87), so conversion won't cannibalize revenue.

Loyalty Dominates

3,116 loyal customers form the core base, with 701 returning and only 83 new — indicating strong retention but weak acquisition.

Product & Category Performance

Top-Rated Products

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80

Top Products by Category

Category	#1 Product	Orders
Clothing	Blouse / Pants	171
Accessories	Jewelry	171
Footwear	Sandals	160
Outerwear	Jacket	163



Discount Dependency & Repeat Buyers

Discount-Heavy Products

Hats (50%), Sneakers (49.7%), Coats (49.1%) — nearly half of purchases driven by discounts, risking margin erosion.

Repeat Buyer Behaviour

2,518 non-subscribers are repeat buyers vs. 958 subscribers — loyalty exists outside subscription, suggesting untapped conversion potential.

Power BI Dashboard



3.9K Customers

Stable, moderately active user base

Clothing Leads

~\$100K revenue, highest across all categories

Young Adults #1

~\$60K revenue, top age group by spend

3.75 Avg. Rating

Satisfied customers, but room to improve

Key Insights & Recommendations



Boost Subscriptions

Only 27% subscribed — promote exclusive benefits to convert the 73% non-subscriber base.



Reward Loyalty

Strengthen retention programs targeting young adults and move returning buyers into the loyal segment.



Review Discount Policy

Balance promotional discounts with margin control — especially for hats, sneakers, and coats.



Expand Weak Categories

Invest in footwear and outerwear to reduce revenue concentration in clothing and accessories.

